

DAY-2 POWER SUPPLY SYSTEM

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

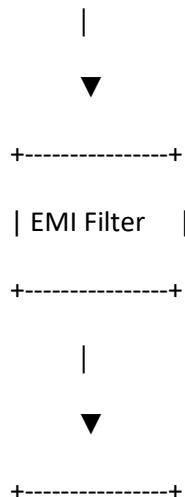
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Device	Type	Why It Needs This Power Supply
Desktop Computer	SMPS	Converts AC power from the wall into DC power required by computer components.
LED Monitor	SMPS	Provides efficient and stable power for the display electronics.
Wi-Fi Router	SMPS	Supplies regulated DC power while consuming less electricity.
Desktop Computer	UPS	Provides backup power during power outages to prevent data loss and sudden shutdowns.
Network Router/Modem	UPS	Keeps the internet connection running during short power interruptions.

1. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

=> AC INPUT (230V AC)



| Rectifier |

| (AC → DC) |

+-----+

|



+-----+

| Filter Capacitor |

+-----+

|



+-----+

| Switching |

| Transistor |

+-----+

|



+-----+

| Transformer |

+-----+

|



+-----+

| Rectifier & |

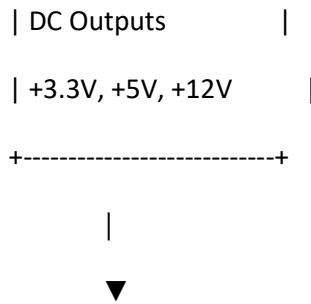
| Filter (DC) |

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Motherboard, CPU, GPU, HDD/SSD

- Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

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- **ATX 24-pin Connector** – The main power connector that supplies power from the SMPS to the motherboard.
- **SATA Power Connector** – Commonly used for modern HDDs, SSDs, and DVD drives.
- **Molex Connector** – An older power connector still found in some fans and legacy devices.

PCIe 6/8-pin Connector – Provides additional power required by high-performance graphics cards.

- Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

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SMPS

